Part B: Multimodal LLM-Based Feature Engineering

MGMT 687 – AI for Business Decisions | Team Project

Purpose: Extract a rich, structured feature set from product images and titles using a multimodal LLM (Groq / LLaMA-4-Scout).
In addition to LLM-extracted features, image statistics (brightness, contrast) are computed with PIL and text/image embeddings are generated locally with `sentence-transformers`.
All features are exported as `features_train.csv` and `features_test.csv` for use in Parts C and D.

Outputs: `features_train.csv`, `features_test.csv`

```
# ── Install Dependencies ──────────────────────────────────────────────
!pip install groq pillow requests sentence-transformers -q
# sentence-transformers includes CLIP support for image embeddings
```

```
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```

```
# ── Imports ──────────────────────────────────────────────────────────
import pandas as pd
import numpy as np
import requests
import base64
import json
import time
import re
import os
import warnings
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import seaborn as sns
from io import BytesIO
from PIL import Image, ImageStat
from groq import Groq
from sentence_transformers import SentenceTransformer

warnings.filterwarnings('ignore')
plt.rcParams.update({'figure.dpi': 130, 'font.size': 11})
print('All libraries loaded successfully.')
```

```
All libraries loaded successfully.
```

Step 1: Load Datasets

```
# ── Upload and Load Data ──────────────────────────────────────────────
# Upload dataset_history.csv and dataset_new.csv when prompted
from google.colab import files
print('Please upload: dataset_history.csv and dataset_new.csv')
uploaded = files.upload()

df_train = pd.read_csv('dataset_history.csv')
df_test  = pd.read_csv('dataset_new.csv')

print(f'Training set  : {df_train.shape[0]} rows x {df_train.shape[1]} cols')
print(f'Test set      : {df_test.shape[0]}  rows x {df_test.shape[1]}  cols')
print(f"Purchase rate : {df_train['ordered'].mean():.1%}  ({df_train['ordered']
df_train.head(3)
```

```
Please upload: dataset_history.csv and dataset_new.csv
```

Choose files 2 files

dataset_history.csv(text/csv) - 61723 bytes, last modified: 05/03/2026 - 100% done

dataset_new.csv(text/csv) - 937 bytes, last modified: 05/03/2026 - 100% done

```
Saving dataset_history.csv to dataset_history (1).csv
Saving dataset_new.csv to dataset_new (1).csv
Training set  : 331 rows x 5 cols
Test set      : 5  rows x 4  cols
Purchase rate : 31.7%  (105 ordered / 331 total)
```

	title	image	price	ordered	image_path
0	Men's Cotton Linen Casual Long Sleeve Shirt Li...	http://t3.gstatic.com/shopping?q=tbn:ANd9GcSwp...	35.44	1	image\image_0.jpg
1	Casual Checked Flannel Shirt White/Red / L	http://t0.gstatic.com/shopping?q=tbn:ANd9GcR7c...	29.70	1	image\image_1.jpg
2	Men's Linen Long Sleeve Shirts Khaki / 5XL	http://t3.gstatic.com/shopping?q=tbn:ANd9GcQld...	34.44	0	image\image_2.jpg

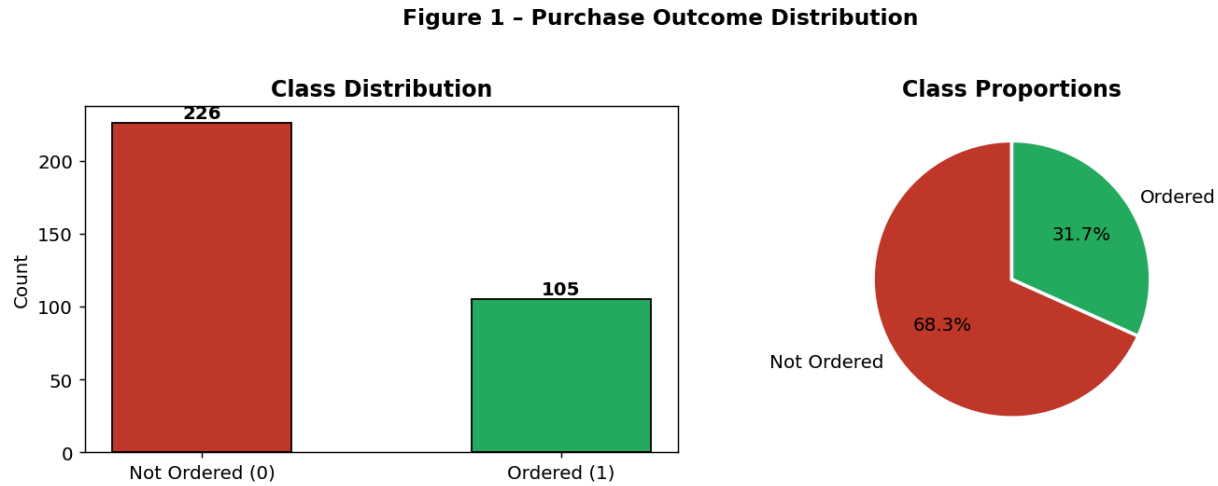
Next steps: Generate code with `df_train` New interactive sheet

```
# ── Class Balance Visualisation ──────────────────────────────────────
fig, axes = plt.subplots(1, 2, figsize=(11, 4))

counts = df_train['ordered'].value_counts().sort_index()
axes[0].bar(['Not Ordered (0)', 'Ordered (1)'], counts.values,
            color=['#c0392b','#27ae60'], edgecolor='black', width=0.5)
axes[0].set_title('Class Distribution', fontweight='bold')
axes[0].set_ylabel('Count')
for i, v in enumerate(counts.values):
    axes[0].text(i, v + 3, str(v), ha='center', fontweight='bold')

axes[1].pie(counts.values, labels=['Not Ordered','Ordered'],
            colors=['#c0392b','#27ae60'], autopct='%1.1f%%',
            startangle=90, wedgeprops={'edgecolor':'white','linewidth':2})
axes[1].set_title('Class Proportions', fontweight='bold')
```

```
plt.suptitle('Figure 1 – Purchase Outcome Distribution', fontsize=13, fontweigh
plt.tight_layout()
plt.savefig('fig1_class_balance.png', dpi=150, bbox_inches='tight')
plt.show()
print(f'Class imbalance ratio  1:0 = 1:{counts[0]/counts[1]:.1f}')
```




```
Class imbalance ratio  1:0 = 1:2.2
```

Step 2: Title Parsing (Size, Price Tier)

Several structured features can be reliably extracted from the product title using regular expressions,
avoiding API usage for straightforward pattern matching.

```
# ── Title Parsing: Size and Price Tier ──────────────────────────────
SIZE_MAP = {
    'XS':0, 'S':1, 'M':2, 'L':3, 'XL':4,
    '2XL':5, 'XXL':5, '3XL':6, 'XXXL':6,
    '4XL':7, '5XL':8, '6XL':9
}

def parse_size(title):
    """Extract size label from title string."""
    # Matches patterns like '/ L', '/ 4XL', '/M', etc. (common e-commerce forma
    match = re.search(r'/\s*([2-9]?XL|XXXL?|[SMLX]{1,3}S?|\d+XL)\b', str(title)
    if match:
        return match.group(1).upper().replace('XXL','2XL').replace('XXXL','3XL'
    return 'Unknown'

def size_numeric(label):
    return SIZE_MAP.get(label, -1)

def size_category(label):
```

```
    idx = SIZE_MAP.get(label, -1)
    if idx <= 1:  return 'small'
    elif idx <= 3: return 'medium'
    elif idx <= 4: return 'large'
    elif idx >= 5: return 'plus'
    return 'unknown'

def price_tier(price):
    """Assign price tier based on training distribution quartiles."""
    if   price < 20:  return 'budget'
    elif price < 35:  return 'mid_range'
    elif price < 60:  return 'premium'
    else:             return 'luxury'

# Apply to both datasets
for df in [df_train, df_test]:
    df['size_label']        = df['title'].apply(parse_size)
    df['size_numeric_index']= df['size_label'].apply(size_numeric)
    df['size_category']     = df['size_label'].apply(size_category)
    df['price_tier']        = df['price'].apply(price_tier)

print('Title parsing complete.')
print(df_train[['title','size_label','size_numeric_index','size_category','pric
```

```
Title parsing complete.
                                                        title size_label  size_nu
Men's Cotton Linen Casual Long Sleeve Shirt Light Blue / 4XL        4XL
                  Casual Checked Flannel Shirt White/Red / L          L
                  Men's Linen Long Sleeve Shirts Khaki / 5XL        5XL
                     Leisure Textured Shirt Set Dark Red / L          L
      Men's Premium Washed Cotton Pocket Shirt Dark Grey / L          L
```

Step 3: Groq API Setup

> **Instructions:** Replace `YOUR_GROQ_API_KEY` with your key from https://console.groq.com/keys
> Free tier supports ~~30 requests/minute. The notebook processes ~336 products with a 2.5 s delay~~ (14 minutes).
> **Progress is saved every 10 rows** — safe to interrupt and resume.

```
# ── API Configuration ──────────────────────────────────────────────
GROQ_API_KEY  = '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'
VISION_MODEL  = 'meta-llama/llama-4-scout-17b-16e-instruct'  # multimodal, Groc
REQUEST_DELAY = 2.5   # seconds between calls — keep ≤ 30 RPM
```

```
client = Groq(api_key=GROQ_API_KEY)

# Connectivity test
try:
    _test = client.chat.completions.create(
        model=VISION_MODEL,
        messages=[{'role':'user','content':'Reply with only: OK'}],
        max_tokens=5
    )
    print('Groq API connection: ✓', _test.choices[0].message.content.strip())
except Exception as e:
    print(f'Connection error: {e}\nVerify API key and model name.')
```

```
Groq API connection: ✓ OK
```

Step 4: Prompt Design and Feature Extraction Function

Prompt Design Rationale

A single **combined zero-shot extraction** prompt is used per product, covering both:

1. **Structural features** – product type, material, color, fit, season, etc.
2. **Subjective quality scores** (1–10) – style appeal, trend alignment, color vibrancy, etc.

Combining both in one call minimises API usage and ensures consistent context.
An expert *persona* anchors the model to fashion merchandising judgment.
A **fixed JSON schema** with explicit value constraints ensures parseable, consistent output.

```
# ── System Prompt ──────────────────────────────────────────────────
SYSTEM_PROMPT = (
    'You are an expert e-commerce fashion merchandising analyst with 15 years c
    'evaluating apparel for major online retailers. You assess products for com
    'based on visual presentation, material quality, style trends, and target m
    'You respond ONLY with valid JSON — no markdown, no commentary.'
)

# ── User Prompt Template ───────────────────────────────────────────
USER_PROMPT_TEMPLATE = '''Analyze this apparel product from the image and title
Product title: "{title}"

Return ONLY a valid JSON object with these exact keys and value constraints:
{{
  "product_category": <"tops"|"bottoms"|"outerwear"|"footwear"|"accessories">,
```

```
  "product_type": <specific: "shirt"|"t-shirt"|"polo"|"jacket"|"pants"|"shorts"
  "gender": <"men"|"women"|"unisex">,
  "primary_material": <dominant fabric: "cotton"|"linen"|"polyester"|"wool"|"si
  "secondary_material": <secondary fabric string or null>,
  "material_category": <"natural"|"synthetic"|"blend">,
  "primary_color": <dominant color, lowercase string>,
  "secondary_color": <second color or null>,
  "pattern_type": <"solid"|"striped"|"checked"|"floral"|"geometric"|"abstract"|
  "sleeve_length": <"sleeveless"|"short_sleeve"|"long_sleeve"|"three_quarter"|"
  "collar_type": <"crew"|"v_neck"|"polo"|"button_down"|"turtleneck"|"hood"|"sta
  "fit_type": <"slim"|"regular"|"relaxed"|"oversized">,
  "fashion_style": <"casual"|"formal"|"streetwear"|"athleisure"|"classic"|"bohe
  "has_buttons": <true or false>,
  "button_count": <integer 0 if none>,
  "has_chest_pocket": <true or false>,
  "pocket_count": <total visible pockets integer>,
  "season_category": <"spring"|"summer"|"fall"|"winter"|"all_season">,
  "background_type": <"white_studio"|"colored_studio"|"natural_setting"|"on_moc
  "style_appeal": <integer 1-10: overall visual attractiveness>,
  "trend_alignment": <integer 1-10: alignment with current fashion trends>,
  "color_vibrancy": <integer 1-10: color attractiveness and commercial appeal>,
  "quality_perception": <integer 1-10: perceived material and construction qual
  "uniqueness_score": <integer 1-10: differentiation from generic products>,
  "versatility_score": <integer 1-10: suitability across multiple occasions>,
  "visual_complexity": <integer 1-10: 1=minimal/plain, 10=highly detailed/patte
  "target_demographic": <"young_casual"|"professional"|"mature_casual"|"athleti
  "occasion_type": <"casual"|"formal"|"both"|"sporty">,
  "is_seasonal": <true if clearly season-specific, false otherwise>
}}'''

print(f'Prompt template defined — extracting 29 features per product in a singl
```

```
Prompt template defined — extracting 29 features per product in a single API cal
```

```
# ── Image Utility and Extraction Function ──────────────────────────────
def fetch_image_as_base64(url, timeout=10):
    """Download image from URL, normalise to JPEG, return (base64_str, mime_typ
    try:
        resp = requests.get(url, timeout=timeout,
                            headers={'User-Agent': 'Mozilla/5.0'})
        if resp.status_code == 200:
            img = Image.open(BytesIO(resp.content)).convert('RGB')
            buf = BytesIO()
            img.save(buf, format='JPEG')
            b64 = base64.b64encode(buf.getvalue()).decode('utf-8')
            return b64, 'image/jpeg'
    except Exception:
        pass
    return None, None
```

```
def extract_llm_features(title, image_url, max_retries=3):
    """Call Groq multimodal LLM to extract all structured features for one proc
    prompt_text = USER_PROMPT_TEMPLATE.format(title=title)
    img_b64, mime = fetch_image_as_base64(image_url)

    for attempt in range(max_retries):
        try:
            if img_b64:
                content = [
                    {'type': 'image_url',
                     'image_url': {'url': f'data:{mime}; base64,{img_b64}'}},
                    {'type': 'text', 'text': prompt_text}
                ]
            else:
                content = [{'type': 'text',
                            'text': f'No image available. Use title only.\n{prc

            resp = client.chat.completions.create(
                model=VISION_MODEL,
                messages=[
                    {'role': 'system', 'content': SYSTEM_PROMPT},
                    {'role': 'user',   'content': content}
                ],
                max_tokens=600,
                temperature=0.1
            )
            raw = resp.choices[0].message.content.strip()
            # Strip markdown fences if present
            raw = re.sub(r'^```(?:json)?\s*', '', raw, flags=re.MULTILINE)
            raw = re.sub(r'\s*```$', '', raw, flags=re.MULTILINE)
            match = re.search(r'\{.*\}', raw, re.DOTALL)
            if match:
                parsed = json.loads(match.group())
                return parsed, (img_b64 is not None)

        except json.JSONDecodeError as e:
            print(f'    JSON parse error attempt {attempt+1}: {e}')
        except Exception as e:
            wait = 2 ** attempt
            print(f'    Retry {attempt+1} ({wait}s) — {type(e).__name__}: {e}')
            time.sleep(wait)

    return None, False


def default_llm_features():
    """Fallback defaults if extraction fails entirely."""
    return {
        'product_category': 'tops', 'product_type': 'shirt',
```

```
            'gender': 'men', 'primary_material': 'cotton',
            'secondary_material': None, 'material_category': 'natural',
            'primary_color': 'unknown', 'secondary_color': None,
            'pattern_type': 'solid', 'sleeve_length': 'long_sleeve',
            'collar_type': 'button_down', 'fit_type': 'regular',
            'fashion_style': 'casual', 'has_buttons': True,
            'button_count': 5, 'has_chest_pocket': False, 'pocket_count': 0,
            'season_category': 'all_season', 'background_type': 'white_studio',
            'style_appeal': 5, 'trend_alignment': 5, 'color_vibrancy': 5,
            'quality_perception': 5, 'uniqueness_score': 5,
            'versatility_score': 5, 'visual_complexity': 5,
            'target_demographic': 'mature_casual', 'occasion_type': 'casual',
            'is_seasonal': False
        }

print('Extraction functions ready.')
```

```
Extraction functions ready.
```

Step 5: Feature Extraction Pipeline

Progress is checkpoint-saved every 10 rows.
If the session is interrupted, simply re-run this cell — it will resume from the last checkpoint.

```
# ── Dataset Processing Function ──────────────────────────────────────
SAVE_TRAIN = 'features_train_raw.csv'
SAVE_TEST  = 'features_test_raw.csv'

def process_dataset(df, save_path, label='dataset'):
    """Run LLM extraction on all rows with resume support."""
    all_records = []
    start_idx   = 0

    if os.path.exists(save_path):
        prev = pd.read_csv(save_path)
        start_idx   = len(prev)
        all_records = prev.to_dict('records')
        print(f'[Resume] {start_idx} rows already processed.')

    img_ok = 0
    total  = len(df)
    for pos, (i, row) in enumerate(df.iloc[start_idx:].iterrows()):
        title     = str(row['title'])
        image_url = str(row['image'])
        price     = float(row['price'])
```

```
            ordered   = row.get('ordered', np.nan)

            print(f'[{start_idx+pos+1:03d}/{total}] {title[:55]}...', end=' ')
            feats, used_img = extract_llm_features(title, image_url)

            if feats is None:
                feats = default_llm_features()
                status = '⚠ default'
            else:
                img_ok += 1
                status = '🖼 img+txt' if used_img else '📝 txt-only'
            print(status)

            record = {
                'row_idx':   i,
                'title':     title,
                'price':     price,
                'orders':    ordered,          # binary label (NaN for test)
                'image_url': image_url,
                # title-parsed features (already in df)
                'size_label':         row.get('size_label','Unknown'),
                'size_numeric_index': row.get('size_numeric_index', -1),
                'size_category':      row.get('size_category','unknown'),
                'price_tier':         row.get('price_tier','mid_range'),
                **feats
            }
            all_records.append(record)

            if len(all_records) % 10 == 0:
                pd.DataFrame(all_records).to_csv(save_path, index=False)
                print(f'  ✅ Checkpoint saved ({len(all_records)} rows)')

            time.sleep(REQUEST_DELAY)

        result = pd.DataFrame(all_records)
        result.to_csv(save_path, index=False)
        print(f'\n✅ Extraction complete — {len(result)} rows | Image successes: {
        return result
```

```
# ── Run Feature Extraction: Training Set ──────────────────────────────
print('Starting LLM feature extraction on TRAINING set...')
print(f'Estimated time: ~{len(df_train)*REQUEST_DELAY/60:.0f} minutes at {REQUE
df_feat_train_raw = process_dataset(df_train, SAVE_TRAIN, 'training')
df_feat_train_raw.head(3)
```

```
Starting LLM feature extraction on TRAINING set...
Estimated time: ~14 minutes at 2.5s per call
[001/331] Men's Cotton Linen Casual Long Sleeve Shirt Light Blue ... 🖼 img+tx
[002/331] Casual Checked Flannel Shirt White/Red / L... 🖼 img+txt
[003/331] Men's Linen Long Sleeve Shirts Khaki / 5XL... 🖼 img+txt
[004/331] Leisure Textured Shirt Set Dark Red / L... 🖼 img+txt
[005/331] Men's Premium Washed Cotton Pocket Shirt Dark Grey / L... 🖼 img+txt
[006/331] Men's Vintage Premium Washed Shirt Pink / L... 🖼 img+txt
[007/331] Western Cowboy Embroidered Button Down Cotton Shirt (US... 🖼 img+tx
[008/331] Men's Premium Washed Cotton Pocket Shirt White / 2XL... 🖼 img+txt
[009/331] Men's Vintage Premium Washed Shirt Sky Blue / S... 🖼 img+txt
[010/331] Modern Soft Cable Knit Shirt Green / M... 🖼 img+txt
  ✅ Checkpoint saved (10 rows)
[011/331] Men's Cotton Long Sleeve Shirt Khaki / L... 🖼 img+txt
[012/331] Men's Cotton And Linen Classic Casual Stand-up Collar S... 🖼 img+tx
[013/331] Casual Loose Fit Shirt Sets (US Only) White / L... 🖼 img+txt
[014/331] Casual Cotton and Linen Shirt Set White / L... 🖼 img+txt
[015/331] Casual Loose Fit Shirt Sets (US Only) Clear Blue / XL... 🖼 img+txt
[016/331] Casual Loose Fit Shirt Sets (US Only) Purple / L... 🖼 img+txt
[017/331] Men's Vintage Premium Washed Shirt Black / L... 🖼 img+txt
[018/331] Athleisure Stretch Polo Shirt Sets Green / M... 🖼 img+txt
[019/331] 100% Cotton Henley Shirt and Shorts Set Coffee / L... 🖼 img+txt
[020/331] Men's Summer Linen Shirt Pink / M... 🖼 img+txt
  ✅ Checkpoint saved (20 rows)
[021/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[022/331] Men's Premium Washed Cotton Pocket Shirt Pink / L... 🖼 img+txt
[023/331] Classic Wrinkle-Free Dress Shirt (US Only) Black / L... 🖼 img+txt
[024/331] Long Sleeve Button Down Flannel Shirt (US Only) Army Gr... 🖼 img+tx
[025/331] Men's Linen Long Sleeve Shirts Blue / 5XL... 🖼 img+txt
[026/331] Essential Muscle Fit Dress Shirt Burgundy / XL... 🖼 img+txt
[027/331] Long Sleeve Button Down Flannel Shirt (US Only) Black /... 🖼 img+tx
[028/331] Casual Loose Fit Shirt Sets (US Only) Khaki / XXL... 🖼 img+txt
[029/331] Casual Linen Skin-friendly Shirts (US Only) Dark Grey /... 🖼 img+tx
[030/331] Short Sleeve Collarless Golf Polo Shirts (US Only) Red ... 🖼 img+tx
  ✅ Checkpoint saved (30 rows)
[031/331] Casual Loose Fit Shirt Sets (US Only) Dark Green / L... 🖼 img+txt
[032/331] Leisure Textured Shirt Set Red / M... 🖼 img+txt
[033/331] Men's Ice Silk Shirt Thin Breathable Shirt White / 4XL... 🖼 img+txt
[034/331] Long Sleeve Button Down Flannel Shirt (US Only) Wine Re... 🖼 img+tx
[035/331] Trendy Breathable T-shirt Set White / XXXL... 🖼 img+txt
[036/331] Casual Loose Fit Shirt Sets (US Only) Pink / L... 🖼 img+txt
[037/331] Essential Muscle Fit Dress Shirt White / M... 🖼 img+txt
[038/331] Men's Vintage Linen Long Sleeve Shirt Navy Blue / L... 🖼 img+txt
[039/331] Cotton Hooded Shirt Shorts Sets Blue / L... 🖼 img+txt
[040/331] Casual Loose Fit Pure Cotton Sports T-shirt/ Sports Sho... 🖼 img+tx
  ✅ Checkpoint saved (40 rows)
[041/331] Men's Business Casual Checkered Colorblock Short Sleeve... 🖼 img+tx
[042/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[043/331] Casual Wrinkle Free Stretch Shirt (US Only) Blue / L... 🖼 img+txt
[044/331] Men's Linen Long Sleeve Shirts Green / 5XL... 🖼 img+txt
[045/331] Easygoing Textured Shirt Shorts Set Beige / L... 🖼 img+txt
[046/331] Essential Muscle Fit Dress Shirt Navy blue / M... 🖼 img+txt
[047/331] Casual Cotton Linen Shirt Set Navy Blue / XXL... 🖼 img+txt
[048/331] Casual Cotton and Linen Shirt Set Sky Blue / S... 🖼 img+txt
[049/331] Men's Vintage Linen Long Sleeve Shirt Coffee / L... 🖼 img+txt
[050/331] Casual Cotton Linen Shirt Set Navy Blue / XL... 🖼 img+txt
```

```
  ✅ Checkpoint saved (50 rows)
[051/331] Subtle Muscle Fit Dress Shirt White / M... 🖼 img+txt
[052/331] Men's Short Sleeve Pocket Shirt Blue / L... 🖼 img+txt
[053/331] Men's Summer Solid Color Comfortable V-neck Short-sleev... 🖼 img+tx
[054/331] Men's Cotton Long Sleeve Shirt Purple / 5XL... 🖼 img+txt
[055/331] Casual Loose Fit Shirt Sets (US Only) Khaki / XL... 🖼 img+txt
[056/331] Effortless Comfort Textured Shirt Set Khaki / L... 🖼 img+txt
[057/331] Cotton Linen Style Stripe Shirts Blue / L... 🖼 img+txt
[058/331] Casual Graphic Cotton Linen Shirt PAT2 / L... 🖼 img+txt
[059/331] Men's Vintage Linen Long Sleeve Shirt Light Blue / M... 🖼 img+txt
[060/331] Casual Graphic Cotton Linen Shirt PAT1 / L... 🖼 img+txt
  ✅ Checkpoint saved (60 rows)
[061/331] Men's Vintage Linen Long Sleeve Shirt Light Blue / L... 🖼 img+txt
[062/331] Men's Ice Silk Shirt Thin Breathable Shirt White / 5XL... 🖼 img+txt
[063/331] Essential Muscle Fit Dress Shirt White / L... 🖼 img+txt
[064/331] Men's Washed Distressed Cotton Daily Short-sleeved T-sh... 🖼 img+tx
[065/331] Men's Summer Linen Shirt Purple / M... 🖼 img+txt
[066/331] Vintage Knit Button Up Shirt (US Only) White / L... 🖼 img+txt
[067/331] Leisure Textured Shirt Set Dark Green / M... 🖼 img+txt
[068/331] Men's Summer Linen Shirt Brown / L... 🖼 img+txt
[069/331] Short Sleeve Collarless Golf Polo Shirts (US Only) Roya... 🖼 img+tx
[070/331] Vintage-Inspired Geometric Knit Shirt Light Khaki / XXX... 🖼 img+tx
  ✅ Checkpoint saved (70 rows)
[071/331] Everyday Comfort Colorblock Waffle Shirt Black / L... 🖼 img+txt
[072/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[073/331] Classic Wrinkle-Free Dress Shirt (US Only) Army Green /... 🖼 img+tx
[074/331] Short Sleeve Collarless Golf Polo Shirts (US Only) Navy... 🖼 img+tx
[075/331] Casual Linen Short Sleeve Shirt Sets (US Only) Blue / L... 🖼 img+tx
[076/331] Vintage-Inspired Geometric Knit Shirt Green / L... 🖼 img+txt
[077/331] Leisure Textured Shirt Set White / XXXL... 🖼 img+txt
[078/331] Casual Linen Style Henley Shirt (US Only) Cream / M... 🖼 img+txt
[079/331] Men's Ice Silk Shirt Thin Breathable Shirt White / M... 🖼 img+txt
[080/331] Casual Linen Short Sleeve Shirt Sets (US Only) Light Bl... 🖼 img+tx
  ✅ Checkpoint saved (80 rows)
[081/331] Effortless Comfort Textured Shirt Set Navy Blue / L... 🖼 img+txt
[082/331] Men's Vintage Premium Washed Shirt Khaki / L... 🖼 img+txt
[083/331] Men's Vintage Velvet Shirt Black / M... 🖼 img+txt
[084/331] Men's Vintage Linen Long Sleeve Shirt Khaki / M... 🖼 img+txt
[085/331] Casual Loose Fit Shirt Sets (US Only) Light Gr een / L... 🖼 img+txt
[086/331] Men's Cotton printed shirt Blue / M... 🖼 img+txt
[087/331] Men's Washed Distressed Cotton Daily Short-sleeved T-sh... 🖼 img+tx
[088/331] Men's Vintage Premium Washed Shirt Sky Blue / 4XL... 🖼 img+txt
[089/331] Fake Two-Piece T-shirt Shorts Set Khaki / L... 🖼 img+txt
[090/331] Casual Loose Fit Shirt Sets (US Only) Beige / M... 🖼 img+txt
  ✅ Checkpoint saved (90 rows)
[091/331] Leisure Textured Shirt Set White / M... 🖼 img+txt
[092/331] Subtle Muscle Fit Dress Shirt Black / M... 🖼 img+txt
[093/331] Vintage Embroidered Western Cowboy Shirt (US Only) Navy... 🖼 img+tx
[094/331] Men's Vintage Linen Long Sleeve Shirt Coffee / M... 🖼 img+txt
[095/331] Leisure Textured Shirt Set Red / XXXL... 🖼 img+txt
[096/331] Western Cowboy Embroidered Button Down Cotton Shirt (US... 🖼 img+tx
[097/331] Casual Cotton Linen Shirt Set Navy Blue / L... 🖼 img+txt
[098/331] Classic Casual Button Down Cotton Linen Shirt (Us Only)... 🖼 img+tx
[099/331] Chictango Striped Henley Shirt Apricot / L... 🖼 img+txt
[100/331] Casual Loose Fit Shirt Sets (US Only) White / XL... 🖼 img+txt
  ✅ Checkpoint saved (100 rows)
[101/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
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[101/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[102/331] Classic Casual Printed Button Down Cotton Linen Shirt (... 🖼 img+tx
[103/331] Vintage Shirt Coat (US Only) Dark Red / L... 🖼 img+txt
[104/331] Men's Comfort Vintage Cotton Long Sleeve Basic T-Shirt ... 🖼 img+tx
[105/331] Short Sleeve Collarless Golf Polo Shirts (US Only) Red ... 🖼 img+tx
[106/331] Men's Casual Corduroy Pocket Short Sleeve Shirt Brown /... 🖼 img+tx
[107/331] Men's Vintage Premium Washed Shirt Khaki / 5XL... 🖼 img+txt
[108/331] Men's Vintage Premium Washed Shirt White / L... 🖼 img+txt
[109/331] Casual Loose Fit Shirt Sets (US Only) Black / M... 🖼 img+txt
[110/331] Men's Washed Distressed Cotton Daily Short-sleeved T-sh... 🖼 img+tx
  ✅ Checkpoint saved (110 rows)
[111/331] Vintage-Inspired Geometric Knit Shirt Black / L... 🖼 img+txt
[112/331] Men's Vintage Linen Long Sleeve Shirt Yellow / M... 🖼 img+txt
[113/331] Casual Loose Fit Shirt Sets (US Only) Purple / M... 🖼 img+txt
[114/331] Casual Loose Fit Pure Cotton Sports T-shirt/ Sports Sho... 🖼 img+tx
[115/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[116/331] Short Sleve Collarless Golf Polo Shirts (US Only) Blue ... 🖼 img+tx
[117/331] Classic Casual Printed Button Down Cotton Linen Shirt (... 🖼 img+tx
[118/331] Men's Vintage Premium Washed Shirt White / M... 🖼 img+txt
[119/331] Casual Linen Blend Guayabera Shirt (US Only) White / L... 🖼 img+txt
[120/331] Essential Muscle Fit Dress Shirt Navy blue / L... 🖼 img+txt
  ✅ Checkpoint saved (120 rows)
[121/331] Men's Vintage Velvet Shirt Wine Red / L... 🖼 img+txt
[122/331] Men's Summer Solid Color Comfortable V-neck Short-sleev... 🖼 img+tx
[123/331] Casual Linen Short Sleeve Shirt Sets (US Only) Khaki / ... 🖼 img+tx
[124/331] Essential Muscle Fit Dress Shirt Black / L... 🖼 img+txt
[125/331] Men's Outdoor Casual Shirt Purple / L... 🖼 img+txt
[126/331] Men's Short Sleeve Pocket Shirt Blue / 5XL... 🖼 img+txt
[127/331] Striped Casual Shirt Blue / M... 🖼 img+txt
[128/331] Cotton Linen Style Henley Shirt (US Only) Khaki / L... 🖼 img+txt
[129/331] Vintage Embroidered Western Cowboy Shirt (US Only) Army... 🖼 img+tx
[130/331] Modern Soft Cable Knit Shirt Black / L... 🖼 img+txt
  ✅ Checkpoint saved (130 rows)
[131/331] Modern Soft Cable Knit Shirt Green / XL... 🖼 img+txt
[132/331] Men's Vintage Premium Washed Shirt Sky Blue / 2XL... 🖼 img+txt
[133/331] Casual Cotton Hollow Lace Shirt Set Pink / L... 🖼 img+txt
[134/331] Embroidered Cowboy Button Down Shirt (US Only) White / ... 🖼 img+tx
[135/331] Casual Cotton Linen Shirt Set White / M... 🖼 img+txt
[136/331] Classic Casual Button Down Cotton Linen Shirt (Us Only)... 🖼 img+tx
[137/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[138/331] Casual Linen Blend Guayabera Shirt (US Only) Dark Red /... 🖼 img+tx
[139/331] Embroidered Cowboy Button Down Shirt (US Only) Coffee /... 🖼 img+tx
[140/331] Men's Linen Striped Half Button Shirt Mustard Yellow / ... 🖼 img+tx
  ✅ Checkpoint saved (140 rows)
[141/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[142/331] Men's Cotton printed shirt Purple / L... 🖼 img+txt
[143/331] Men's Linen Henley Long Sleeve Shirt Pink / L... 🖼 img+txt
[144/331] Casual Cotton Linen Shirt Set White / XXL... 🖼 img+txt
[145/331] Men's Ice Silk Shirt Thin Breathable Shirt White / L... 🖼 img+txt
[146/331] Men's Washed Distressed Cotton Daily Short-sleeved T-sh... 🖼 img+tx
[147/331] Essential Muscle Fit Dress Shirt Light Green / L... 🖼 img+txt
[148/331] Men's Premium Washed Cotton Pocket Shirt Khaki / L... 🖼 img+txt
[149/331] Men's Luxurious Touch Stretch Cotton T-Shirt Black / S... 🖼 img+txt
[150/331] Men's Summer Linen Shirt Orange / M... 🖼 img+txt
  ✅ Checkpoint saved (150 rows)
[151/331] Stylish Contrast Waffle Shirt Set Cream / XXL... 🖼 img+txt
[152/331] Short Sleeve Collarless Golf Polo Shirts (US Only) Gree... 🖼 img+tx
```

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[153/331] Men's Linen Henley Long Sleeve Shirt Blue / L... 🖼 img+txt
[154/331] Casual Loose Fit Shirt Sets (US Only) Blue / XXL... 🖼 img+txt
[155/331] Men's Washed Distressed Cotton Daily Short-sleeved T-sh... 🖼 img+tx
[156/331] Casual Wrinkle Free Stretch Shirt (US Only) Navy Blue /... 🖼 img+tx
[157/331] Casual Graphic Cotton Linen Shirt PAT3 / L... 🖼 img+txt
[158/331] Casual Linen Blend Shirt (US Only) Light Khaki / L... 🖼 img+txt
[159/331] Men's Cotton printed shirt Blue / L... 🖼 img+txt
[160/331] Men's Vintage Premium Washed Shirt Blue / M... 🖼 img+txt
  ✅ Checkpoint saved (160 rows)
[161/331] Cotton Linen Style Stripe Shirts Blue / 5XL... 🖼 img+txt
[162/331] Casual Linen Short Sleeve Shirt Sets (US Only) Light Bl... 🖼 img+tx
[163/331] Vintage Embroidered Western Cowboy Shirt (US Only) Blac... 🖼 img+tx
[164/331] Cotton Linen Style Henley Shirt (US Only) Denim Blue / ... 🖼 img+tx
[165/331] Men's Fashion Casual Pleated Fabric Short-sleeved Lapel... 🖼 img+tx
[166/331] Breathable Button Down Knit Shurt Army Green / L... 🖼 img+txt
[167/331] Cotton Linen Solid Color Double Pockets Shirt Army Gree... 🖼 img+tx
[168/331] Short Sleeve Collarless Golf Pooo Shirts (US Only) Grey... 🖼 img+tx
[169/331] Men's Vintage Premium Washed Shirt Army Green / L... 🖼 img+txt
[170/331] Western Cowboy Style Cotton Shirt (US Only) White / L... 🖼 img+txt
  ✅ Checkpoint saved (170 rows)
[171/331] Essential Muscle Fit Dress Shirt Burgundy / L... 🖼 img+txt
[172/331] Casual Linen Short Sleeve Shirt Sets (US Only) Wine Red... 🖼 img+tx
[173/331] Essential Muscle Fit Dress Shirt Grey / L... 🖼 img+txt
[174/331] Wrinkle Free Button Down Dress Shirt (US Only) Royal Bl... 🖼 img+tx
[175/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[176/331] Men's Vintage Velvet Shirt Wine Red / 3XL... 🖼 img+txt
[177/331] Cotton Linen Style Henley Shirt (Live Only) Khaki / L... 🖼 img+txt
[178/331] Embroidered Cowboy Button Down Shirt (US Only) Black & ... 🖼 img+tx
[179/331] Men's Casual Texture Lapel Comfortable Short-sleeved Po... 🖼 img+tx
[180/331] Vintage Embroidered Western Cowboy Shirt (US Only) Navy... 🖼 img+tx
  ✅ Checkpoint saved (180 rows)
[181/331] Stylish Soft 2-Piece Shirt Set White / XL... 🖼 img+txt
[182/331] Vintage Embroidered Western Cowboy Shirt (US Only) Ligh... 🖼 img+tx
[183/331] Men's Vintage Premium Washed Shirt Sky Blue / M... 🖼 img+txt
[184/331] Classic Casual Button Down Cotton Linen Shirt (Us Only)... 🖼 img+tx
[185/331] Casual Loose Fit Pure Cotton Sports T-shirt/ Sports Sho... 🖼 img+tx
[186/331] Minimal Comfort Polo Shirt Shorts Set White / L... 🖼 img+txt
[187/331] Men's Vintage Premium Washed Shirt White / 5XL... 🖼 img+txt
[188/331] Men's Luxurious Touch Stretch Cotton T-Shirt White / 4X... 🖼 img+tx
[189/331] Men's Striped Sweatshirt Warm Loose Lapel Peplum Long S... 🖼 img+tx
[190/331] Casual 100% Cotton Stand Collar Shirt White / L... 🖼 img+txt
  ✅ Checkpoint saved (190 rows)
[191/331] Minimal Comfort Polo Shirt Shorts Set Khaki / XXXL... 🖼 img+txt
[192/331] Men's Summer Linen Shirt Purple / 5XL... 🖼 img+txt
[193/331] Muscle Fit Shirts (US Only) Brown / L... 🖼 img+txt
[194/331] Wrinkle Free Button Down Dress Shirt (US Only) Dark Gre... 🖼 img+tx
[195/331] Men's Linen Long Sleeve Shirts Purple / M... 🖼 img+txt
[196/331] Men's Linen Long Sleeve Shirts Gray / L... 🖼 img+txt
[197/331] Men's Premium Washed Cotton Pocket Shirt Army Green / 5... 🖼 img+tx
[198/331] Casual Loose Fit Shirt Sets (US Only) Grey / XL... 🖼 img+txt
[199/331] Men's Ice Silk V-Neck T-Shirt Wine Red+White+Tibetan Bl... 🖼 img+tx
[200/331] Men's Vintage Premium Washed Shirt Army Green / 5XL... 🖼 img+txt
  ✅ Checkpoint saved (200 rows)
[201/331] Men's Vintage Premium Washed Shirt Blue / L... 🖼 img+txt
[202/331] Muscle Fit Shirts (US Only) Royal Blue / L... 🖼 img+txt
[203/331] Embroidered Cowboy Button Down Shirt (US Only) Classic ... 🖼 img+tx
[204/331] Men's Solid Color Linen Short Sleeve Lapel Shirt Pink / ... 🖼 img+tx
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[204/331] Men's Solid Color Linen Short Sleeve Lapel Shirt Pink /... 🖼 img+tx
[205/331] Embroidered Cowboy Button Down Shirt (US Only) Coffee /... 🖼 img+tx
[206/331] Men's Vintage Premium Washed Shirt Pink / M... 🖼 img+txt
[207/331] Men's Vintage Linen Long Sleeve Shirt Pink / M... 🖼 img+txt
[208/331] Men's Cotton Knit Striped Polo Shirt Purple / L... 🖼 img+txt
[209/331] Men's Cotton Linen Casual Long Sleeve Shirt Purple / L... 🖼 img+txt
[210/331] Cotton Linen Style Henley Shirt (US Only) White / L... 🖼 img+txt
  ✅ Checkpoint saved (210 rows)
[211/331] 100% Cotton Henley Shirt and Shorts Set Coffee / XL... 🖼 img+txt
[212/331] Men's Luxurious Touch Stretch Cotton T-Shirt White / M... 🖼 img+txt
[213/331] Casual Loose Fit Shirt Sets (US Only) Black / L... 🖼 img+txt
[214/331] Men's Washed Distressed Cotton Daily Short-sleeved T-sh... 🖼 img+tx
[215/331] Chictango atchwork CA1 Overshirt Army Green / L... 🖼 img+txt
[216/331] Men's Vintage Linen Long Sleeve Shirt Pink / L... 🖼 img+txt
[217/331] Men's Vintage Premium Washed Shirt White / XL... 🖼 img+txt
[218/331] Men's Cotton Knit Striped Polo Shirt Khaki / 4XL... 🖼 img+txt
[219/331] Long Sleeve Button Down Shirt Army Green / L... 🖼 img+txt
[220/331] Men's Summer Solid Color Comfortable V-neck Short-sleev... 🖼 img+tx
  ✅ Checkpoint saved (220 rows)
[221/331] Men's Plaid Collar Summer Shirt Purple / 5XL... 🖼 img+txt
[222/331] Casual Loose Fit Shirt Sets (US Only) White / XXL... 🖼 img+txt
[223/331] Men's Solid Color Linen Short Sleeve Lapel Shirt White ... 🖼 img+tx
[224/331] Embroidered Cowboy Button Down Shirt (US Only) Classic ... 🖼 img+tx
[225/331] Men's Ice Silk V-Neck T-Shirt Light Blue+Black+White / ... 🖼 img+tx
[226/331] Men's Casual Texture Lapel Comfortable Short-sleeved Po... 🖼 img+tx
[227/331] Athleisure Stretch Polo Shirt Sets Green / L... 🖼 img+txt
[228/331] Minimal Comfort Polo Shirt Shorts Set Khaki / L... 🖼 img+txt
[229/331] Classic Casual Button Down Cotton Linen Shirt (Us Only)... 🖼 img+tx
[230/331] 100% Cotton Henley Shirt and Shorts Set Blue / L... 🖼 img+txt
  ✅ Checkpoint saved (230 rows)
[231/331] Men's Vintage Linen Long Sleeve Shird Yellow / L... 🖼 img+txt
[232/331] Men's Vintage Linen Long Sleeve Shirt Black / L... 🖼 img+txt
[233/331] Casual Loose Fit Shirt Sets (US Only) Beige / L... 🖼 img+txt
[234/331] Men's Cotton Linen Casual Long Sleeve Shirt Light Blue ... 🖼 img+tx
[235/331] Men's Linen Long Sleeve Shirts Purple / 5XL... 🖼 img+txt
[236/331] Summer Essential Button Down Knit Shirt (US Only) Brown... 🖼 img+tx
[237/331] Casual Cotton and Linen Shirt Set White / M... 🖼 img+txt
[238/331] Men's Summer Solid Color Comfortable V-neck Short-sleev... 🖼 img+tx
[239/331] Men's Vintage Linen Long Sleeve Shirt Navy Blue / M... 🖼 img+txt
[240/331] Casual Loose Fit Shirt Sets (US Only) Khaki / M... 🖼 img+txt
  ✅ Checkpoint saved (240 rows)
[241/331] Vintage Knit Button Up Shirt (US Only) Green / L... 🖼 img+txt
[242/331] Casual Cotton Linen Shirt Set White / L... 🖼 img+txt
[243/331] Men's Vintage Velvet Shirt Black / L... 🖼 img+txt
[244/331] Luxury Dress Shirt (US Only) Beige / L... 🖼 img+txt
[245/331] Vintage-Inspired Geometric Knit Shirt Light Khaki / L... 🖼 img+txt
[246/331] Casual Lcose Fit Shirt Sets (US Only) Clear Blue / L... 🖼 img+txt
[247/331] Men's Cotton Knit Striped Polo Shirt Purple / M... 🖼 img+txt
[248/331] Minimal Comfort Polo Shirt Shorts Set White / M... 🖼 img+txt
[249/331] Casual Loose Fit Shirt Sets (US Only) Blue / L... 🖼 img+txt
[250/331] Men's Luxurious Touch Stretch Cotton T-Shirt Black / M... 🖼 img+txt
  ✅ Checkpoint saved (250 rows)
[251/331] Men's Cotton Linen Short Sleeve Shirt Dark Blue / L... 🖼 img+txt
[252/331] Men's Cotton Long Sleeve Shirt Black / L... 🖼 img+txt
[253/331] Men's Washed Distressed Cotton Daily Short-sleeved T-sh... 🖼 img+tx
[254/331] Men's Summer Linen Shirt Purple / L... 🖼 img+txt
[255/331] Gentleman's Casual Basic Cotton Linen Shirt Purple / L... 🖼 img+txt
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[256/331] Men's Luxurious Touch Stretch Cotton T-Shirt White / 5X... 🖼 img+tx
[257/331] Men's Washed Distressed Cotton Daily Short-sleeved T-sh... 🖼 img+tx
[258/331] Men's Casual Corduroy Pocket Short Sleeve Shirt Black /... 🖼 img+tx
[259/331] Men's Abstract Gradient Geometric T-Shirt Top Purple / ... 🖼 img+tx
[260/331] Casual Linen Blend Guayabera Shirt (US Only) White / M... 🖼 img+txt
  ✅ Checkpoint saved (260 rows)
[261/331] Men's Linen Henley Long Sleeve Shirt Light Grey / M... 🖼 img+txt
[262/331] Men's Fashion Casual Pleated Fabric Short-sleeved Lapel... 🖼 img+tx
[263/331] Men's Fashion Turndown Collar Buttons Striped Shirt Gre... 🖼 img+tx
[264/331] Leisure Textured Shirt Set Dark Green / L... 🖼 img+txt
[265/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[266/331] Casual Loose Fit Shirt Sets (US Only) White / M... 🖼 img+txt
[267/331] Summer Cuban Guayabera Shirt (US Only) Light Green / L... 🖼 img+txt
[268/331] Vintage Knit Button Up Shart (US Only) Navy Blue / L... 🖼 img+txt
[269/331] Men's Summer Linen Shirt Orange / L... 🖼 img+txt
[270/331] Casual Loose Fit Shirt Sets (US Only) Khaki / L... 🖼 img+txt
  ✅ Checkpoint saved (270 rows)
[271/331] Men's Vintage Premium Washeh Shirt Khaki / M... 🖼 img+txt
[272/331] Men's Luxurious Touch Stretch Cotton T-Shirt Black / L... 🖼 img+txt
[273/331] Leisure Textured Shirt Set Red / S... 🖼 img+txt
[274/331] Stylish Slant Placket Tied Shirt Apricot / L... 🖼 img+txt
[275/331] Men's Cotton printed shirt Army Green / L... 🖼 img+txt
[276/331] Muscle Fit Shirts (US Only) Dark Purple / L... 🖼 img+txt
[277/331] Men's Linen Resort CasualhButton-Down Everyday Short Sl... 🖼 img+tx
[278/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[279/331] Men's Summer Solid Color Comfortable V-neck Short-sleev... 🖼 img+tx
[280/331] Cotton Linen Style Stripe Shirts Blue / M... 🖼 img+txt
  ✅ Checkpoint saved (280 rows)
[281/331] Muscle Fit Shirts (US Only) Navy Blue / M... 🖼 img+txt
[282/331] Leisure Textured Shirt Set Red / L... 🖼 img+txt
[283/331] Casual Loose Fit Pure Cotton Sports T-shirt/ Sports Sho... 🖼 img+tx
[284/331] Men's Luxurious Touch Stretch Cotton T-Shirt White / XL... 🖼 img+tx
[285/331] Vintage Shirt Coat (US Only) Purple / L... 🖼 img+txt
[286/331] Casual Loose Fit Shirt Sets (US Only) Grey / L... 🖼 img+txt
[287/331] Men's Summer Solid Color Comfortable V-neck Short-sleev... 🖼 img+tx
[288/331] Men's Cotton Long Sleeve Shirt Purple / L... 🖼 img+txt
[289/331] Vintage-Inspired Geometric Knit Shirt Light Khaki / XL... 🖼 img+txt
[290/331] Essential Muscle Fit Dress Shirt Grey / M... 🖼 img+txt
  ✅ Checkpoint saved (290 rows)
[291/331] Men's Combed Cotton T-Shirt Purple / L... 🖼 img+txt
[292/331] Western Cowboy Embroidered Button Down Cotton Shirt (US... 🖼 img+tx
[293/331] Long Sleeve Cotton Linen Henley Shirt (US Only) Dark Bl... 🖼 img+tx
[294/331] Men's Plaid Collar Summer Shirt Light blue / L... 🖼 img+txt
[295/331] Men's Summer Linen Shirt Pink / L... 🖼 img+txt
[296/331] Men's Washed Distressed Cotton Daily Short-sleeved T-sh... 🖼 img+tx
[297/331] Embroidered Cowboy Button Down Shirt (US Only) Classic ... 🖼 img+tx
[298/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[299/331] Men's Fashion Turndown Collar Buttons Striped Shirt Red... 🖼 img+tx
[300/331] Ribbed Textured Knit Polo Shirt (US Only) Pink / M... 🖼 img+txt
  ✅ Checkpoint saved (300 rows)
[301/331] Men's Short Sleeve Pocket Shirt Dark Khaki / 5XL... 🖼 img+txt
[302/331] Casual Cotton Linen Shirt Set White / XL... 🖼 img+txt
[303/331] Muscle Fit Shirts (rS Only) Red / L... 🖼 img+txt
[304/331] Men's Vintage Velvet Shirt Wine Red / M... 🖼 img+txt
[305/331] Casual Lightweight Henley Shirt (US Only) Navy Blue / X... 🖼 img+tx
[306/331] Men's Vintage Premium Washed Shirt Blue / XL... 🖼 img+txt
[307/331] Rose Printed Slim Fit Dress Shirts (US Only) Black / L... 🖼 img+txt
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[308/331] Western Cowboy Style Cotton Shirt (US Only) Black / L... 🖼 img+txt
[309/331] Cotton Linen Style Stripe Shirts Red / M... 🖼 img+txt
[310/331] Muscle Fit Shirts (US Only) Navy Blue / L... 🖼 img+txt
  ✅ Checkpoint saved (310 rows)
[311/331] Men's Linen Long Sleeve Shirts Purple / L... 🖼 img+txt
[312/331] Modern Soft Cable Knit Shirt Green / L... 🖼 img+txt
[313/331] Soft Cotton Linen Button Shirt (US Only) Green / M... 🖼 img+txt
[314/331] Leisure Textured Shirt Set Pink / XXXL... 🖼 img+txt
[315/331] Casual Linen Skin-friendly Shirts (US Only) Green / L... 🖼 img+txt
[316/331] Classic Wrinkle-Free Dress Shirt (US Only) Brown / L... 🖼 img+txt
[317/331] Casual Linen Skin-friendly Shirts (US Only) Light Blue ... 🖼 img+tx
[318/331] Men's Linen Resort Casual Button-Down Everyday Short Sl... 🖼 img+tx
[319/331] Men's Short Sleeve Pocket Shirt Army Green / 4XL... 🖼 img+txt
[320/331] Vintage Embroidered Western Cowboy Shirt (US Only) Khak... 🖼 img+tx
  ✅ Checkpoint saved (320 rows)
[321/331] Men's Cotton Linen Casual Long Sleeve Shirt Blue / L... 🖼 img+txt
[322/331] Linen Henley Long Sleeve Shirts with Pocket (US Only) B... 🖼 img+tx
[323/331] Cotton Beach Button Down Long Sleeve Shirt (US Only) Wh... 🖼 img+tx
[324/331] Men's Combed Cotton T-Shirt Blue / L... 🖼 img+txt
[325/331] Men's Luxurious Touch Stretch Cotton T-Shirt White / L... 🖼 img+txt
[326/331] Men's Linen Long Sleeve Shirts White / 5XL... 🖼 img+txt
[327/331] Chictango Pink Corduroy Shirt Set Pink / L... 🖼 img+txt
[328/331] Men's Ice Silk V-Neck T-Shirt Black+Grey+Tibetan Blue /... 🖼 img+tx
[329/331] Casual Linen Short Sleeve Shirt Sets (US Only) Light Kh... 🖼 img+tx
[330/331] Classic Casual Button Down Cotton Linen Shirt (Us Only)... 🖼 img+tx
  ✅ Checkpoint saved (330 rows)
[331/331] Casual Cotton Linen Shirt Set White / XXXL... 🖼 img+txt

✅ Extraction complete — 331 rows | Image successes: 331/331
```

	row_idx	title	price	orders	image_url	size_label	si
0	0	Men's Cotton Linen Casual Long Sleeve Shirt Li...	35.44	1	http://t3.gstatic.com/shopping?q=tbn:ANd9GcSwp...	4XL	
1	1	Casual Checked Flannel Shirt White/Red / L	29.70	1	http://t0.gstatic.com/shopping?q=tbn:ANd9GcR7c...	L	
2	2	Men's Linen Long Sleeve Shirts Khaki / 5XL	34.44	0	http://t3.gstatic.com/shopping?q=tbn:ANd9GcQId...	5XL	

3 rows × 38 columns

```
# ── Run Feature Extraction: Test Set ──────────────────────────────────
print('Starting LLM feature extraction on TEST set (5 products)...')
df_feat_test_raw = process_dataset(df_test, SAVE_TEST, 'test')
df_feat_test_raw
```

```
Starting LLM feature extraction on TEST set (5 products)...
[001/5] Men's Vintage Premium Washed Shirt Navy Blue / L... 🖼 img+txt
[002/5] Ribbed Textured Knit Polo Shirt (US Only) Khaki / L... 🖼 img+txt
[003/5] Men's Rose Print Vintage Casual Button Short Sleeve Shi... 🖼 img+txt
[004/5] Classic Casual Button Down Cotton Linen Shirt (Us Only)... 🖼 img+txt
[005/5] Men's Casual Geo Pattern Short Sleeve Shirt & Shorts 2P... 🖼 img+txt

✅ Extraction complete — 5 rows | Image successes: 5/5
```

	row_idx	title	price	orders	image_url	size_label	size_
0	0	Men's Vintage Premium Washed Shirt Navy Blue / L	59.09	NaN	http://t3.gstatic.com/shopping?q=tbn:ANd9GcQFE...	L	
1	1	Ribbed Textured Knit Polo Shirt (US Only) Khak...	35.67	NaN	http://t1.gstatic.com/shopping?q=tbn:ANd9GcTjq...	L	
2	2	Men's Rose Print Vintage Casual Button Short S...	27.88	NaN	http://t1.gstatic.com/shopping?q=tbn:ANd9GcRym...	L	
3	3	Classic Casual Button Down Cotton Linen Shirt ...	35.96	NaN	http://t1.gstatic.com/shopping?q=tbn:ANd9GcTuz...	L	
4	4	Men's Casual Geo Pattern Short Sleeve Shirt & ...	39.16	NaN	http://t1.gstatic.com/shopping?q=tbn:ANd9GcSXE...	L	

5 rows × 38 columns

Step 6: Image Statistics (Brightness & Contrast)

PIL-computed pixel-level statistics are computed independently of the LLM, providing objective, reproducible image quality signals.

```
# ── Compute Image Brightness and Contrast ──────────────────────────
def compute_image_stats(url, timeout=8):
    """Return (brightness, contrast) as floats; (-1,-1) if image unavailable."
    try:
        resp = requests.get(url, timeout=timeout,
                            headers={'User-Agent': 'Mozilla/5.0'})
        if resp.status_code == 200:
            img  = Image.open(BytesIO(resp.content)).convert('L')  # grayscale
            stat = ImageStat.Stat(img)
            brightness = stat.mean[0]          # mean pixel value  0–255
            contrast   = stat.stddev[0]        # std dev of pixels 0–255
            return round(brightness, 3), round(contrast, 3)
    except Exception:
        pass
    return -1.0, -1.0


def add_image_stats(df, cache_path, delay=0.3):
    """Add image_brightness and image_contrast columns with progress saving."""
    cache_col_b = 'image_brightness'
    cache_col_c = 'image_contrast'

    if cache_col_b in df.columns:
        print('Image stats already present — skipping.')
        return df

    brightnessL, contrastL = [], []
    for i, row in df.iterrows():
        b, c = compute_image_stats(str(row['image_url']))
        brightnessL.append(b)
        contrastL.append(c)
        if (i+1) % 50 == 0:
            print(f'  Image stats: {i+1}/{len(df)} processed...')
        time.sleep(delay)

    df[cache_col_b] = brightnessL
    df[cache_col_c] = contrastL
    df.to_csv(cache_path, index=False)
    return df

print('Computing image statistics for training set...')
df_feat_train_raw = add_image_stats(df_feat_train_raw, SAVE_TRAIN)
```

```
print('Computing image statistics for test set...')
df_feat_test_raw  = add_image_stats(df_feat_test_raw, SAVE_TEST)
print('\nImage statistics computed.')
print(df_feat_train_raw[['image_brightness','image_contrast']].describe().rounc
```

```
Computing image statistics for training set...
  Image stats: 50/331 processed...
  Image stats: 100/331 processed...
  Image stats: 150/331 processed...
  Image stats: 200/331 processed...
  Image stats: 250/331 processed...
  Image stats: 300/331 processed...
Computing image statistics for test set...

Image statistics computed.
       image_brightness  image_contrast
count            331.00          331.00
mean             151.52           65.28
std               34.80           19.04
min               54.87           17.42
25%              124.16           51.95
50%              149.25           60.75
75%              181.19           80.70
max              234.34          114.35
```

Step 7: Text and Image Embeddings

Embeddings capture semantic and visual information not conveyed by categorical labels.

- **Title embedding** — `all-MiniLM-L6-v2` (384-dim) encodes product title semantics
- **Image embedding** — `clip-ViT-B-32` (512-dim) encodes visual content via CLIP
- **Multimodal embedding** — L2-normalised concatenation of both (896-dim)

All embeddings are computed locally.

```
# ── Load Embedding Models ──────────────────────────────────────────────
print('Loading sentence-transformers models...')
text_model  = SentenceTransformer('sentence-transformers/all-MiniLM-L6-v2')
# CLIP model for image embeddings
clip_model  = SentenceTransformer('clip-ViT-B-32')
print('Models loaded.')


# ── Text Embeddings ────────────────────────────────────────────────────
def get_title_embeddings(df):
    titles = df['title'].fillna('').tolist()
    vecs   = text_model.encode(titles, batch_size=32, show_progress_bar=True, r
```

```
    return vecs  # shape (N, 384)


# ── Image Embeddings via CLIP ──────────────────────────────────────────
def get_image_embeddings(df):
    imgs = []
    for url in df['image_url']:
        try:
            resp = requests.get(str(url), timeout=8,
                                headers={'User-Agent': 'Mozilla/5.0'})
            if resp.status_code == 200:
                img = Image.open(BytesIO(resp.content)).convert('RGB')
                imgs.append(img)
            else:
                imgs.append(None)
        except Exception:
            imgs.append(None)

    # Replace failed fetches with blank white image
    blank = Image.new('RGB', (224, 224), color=(255, 255, 255))
    imgs  = [img if img is not None else blank for img in imgs]
    vecs  = clip_model.encode(imgs, batch_size=16, show_progress_bar=True, norm
    return vecs  # shape (N, 512)


# ── Multimodal Embedding ───────────────────────────────────────────────
def fuse_embeddings(title_vecs, image_vecs):
    """Concatenate and L2-normalise title and image embeddings."""
    combined = np.concatenate([title_vecs, image_vecs], axis=1)  # (N, 896)
    norms    = np.linalg.norm(combined, axis=1, keepdims=True)
    norms    = np.where(norms == 0, 1, norms)  # avoid div-by-zero
    return combined / norms


def serialize_vec(arr):
    """Serialise numpy array to JSON string for CSV storage."""
    return json.dumps(arr.tolist())


def add_embeddings(df, cache_path):
    """Compute and attach all three embedding columns."""
    if 'title_embedding_vector' in df.columns:
        print('Embeddings already present — skipping.')
        return df

    print('Computing title embeddings...')
    title_vecs = get_title_embeddings(df)
    print('Computing image embeddings via CLIP...')
    image_vecs = get_image_embeddings(df)
    print('Fusing embeddings...')
    mm_vecs    = fuse_embeddings(title_vecs, image_vecs)
```

```
    df['title_embedding_vector']  = [serialize_vec(v) for v in title_vecs]
    df['image_embedding_vector']  = [serialize_vec(v) for v in image_vecs]
    df['multimodal_embedding']    = [serialize_vec(v) for v in mm_vecs]
    df.to_csv(cache_path, index=False)
    print('Embeddings saved.')
    return df


print('\nAdding embeddings to training set...')
df_feat_train_raw = add_embeddings(df_feat_train_raw, SAVE_TRAIN)
print('\nAdding embeddings to test set...')
df_feat_test_raw  = add_embeddings(df_feat_test_raw, SAVE_TEST)
print('\nAll embeddings complete.')
```